

How to Write an Academic Paper in Finance – Module 1

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Overview

Module 1: From Idea to Proposal

- ▶ Learning from a Classic: Fama & French (1992)
 - ▶ From Idea to Proposal
- ▶ Find a Research Idea
- ▶ Write Your Research Proposal

Why Fama-French (1992)?

The Perfect Case Study

Reason	Why It Matters
One of the most cited papers in finance	Everyone knows it
Clear, simple idea	Size + Book-to-Market capture returns
Revolutionary contribution	Killed the CAPM as the sole model
Exemplary structure	Follows Cochrane's "newspaper style"

"Two easily measured variables, size and book-to-market equity, combine to capture the cross-sectional variation in average stock returns." — Fama & French (1992), Abstract

The Gap Between Literature and Reality

Fama-French's Starting Point

Literature (CAPM – 1992 consensus)
“Beta explains everything”

← Fama-French's question:
“is this really true?”

Reality (observed anomalies)

- Size effect (Banz 1981)
- Value effect (Stattman 1980)
- Leverage effect (Bhandari 1988)
- E/P effect (Basu 1983)

The Research Question

- ▶ Observation: The Anomalies did not fit CAMP
- ▶ The Puzzle: CAPM says **only beta** should matter. But these variables all seem to matter. Why?
- ▶ Question:
- ▶ "Since E/P, ME, leverage, and BE/ME are all scaled versions of price, it is reasonable to expect that **some of them are redundant** for describing average returns. Our goal is to evaluate the joint roles of market β , size, E/P, leverage, and book-to-market equity in the cross-section of average returns." — Fama & French (1992), p. 428
- ▶ Their **Core Question**:
 - ▶ Which of these variables truly matter, and which are just proxies for others?

Where Did Fama-French Get Their Idea?

- ▶ They built on prior anomalies literature, but their innovation was:
- ▶ Synthesis: Previous papers each found one anomaly. Fama-French asked: "Which ones are real when we **test them together?**"
- ▶ Methodological insight: Previous tests used portfolios sorted on one variable. Fama-French used **double sorting** to separate correlated effects.
- ▶ Parsimony: They sought the **simplest model** that captures everything.

The “Ah! So that explains it! Moment

"The greatest compliment is 'Ah! So that explains it!'" — Varian

- ▶ Before Fama-French:
 - ▶ Finance profession believed: Beta explains everything
 - ▶ But: Many anomalies kept appearing
 - ▶ Problem: No one knew which anomalies were real vs. redundant

The “Ah! So that explains it! Moment

“Our main result is that for the 1963-1990 period, **size and book-to-market equity capture** the cross-sectional variation in average stock returns associated with size, E/P, book-to-market equity, and leverage.” — Fama & French(1992) p. 450

- ▶ After Fama-French:
 - ▶ "Ah! So size and value capture all those anomalies!"
 - ▶ "Ah! That's why beta seems to work sometimes—it's correlated with size!"

Idea Checklist

Question	Fama-French Answer
Can I state the idea in one sentence?	"Size and book-to-market capture the cross-section of average returns."
Is there a genuine gap between theory and reality?	CAPM says beta only; reality shows many anomalies
Would a non-economist find it interesting?	"Why do some stocks earn higher returns than others?"
Is the question answerable?	Yes—data exists (CRSP, Compustat)
Have I let the idea incubate?	Built on a decade of anomalies research

What Would Their Proposal Have Looked Like?

- ▶ Before the paper, there is a plan
- ▶ We call it a research **proposal** which typically includes the following part:
 1. Problem Statement
 2. Expected Result
 3. Research Method
 4. Key References

Problem Statement

"The Capital Asset Pricing Model (Sharpe, 1964; Lintner, 1965) predicts that market beta is the sole determinant of expected stock returns. However, recent empirical studies have documented several 'anomalies' that contradict this prediction: the size effect (Banz, 1981), the value effect (Stattman, 1980; Rosenberg, Reid, and Lanstein, 1985), the leverage effect (Bhandari, 1988), and the earnings-price ratio effect (Basu, 1983).

These variables are **all scaled versions of price**, suggesting they may be redundant proxies for the same underlying risk factors. **This study will evaluate the joint roles** of beta, size, leverage, E/P, and book-to-market equity to determine which variables truly capture the cross-section of expected returns."

Problem Statement

Element	In Their Proposal
What we know	CAPM predicts beta explains everything
What's puzzling	Many anomalies don't fit CAPM
The gap	No one has tested all together
What we'll do	Jointly evaluate all variables

Expected Results

Hypothetical Fama-French Proposal

- ▶ "We hypothesize that many of these variables are **redundant**. Specifically, we expect that:
 1. **Beta will have little or reduced explanatory power** when tested alongside other variables.
 2. **Some of the current anomaly factors will emerge as the key factors**, absorbing the explanatory power of leverage and E/P.
 3. The combination of these key factors will provide a **parsimonious model** that captures most of the cross-sectional variation in average returns."

Expected Results – The Benefits

Benefit	Explanation
Clarifies your thinking	Forces you to commit to a hypothesis
Guides research design	Helps you design tests that can reject your hypothesis
Shows you understand the literature	Expected results come from theory and prior evidence
Makes the contribution clear	Readers know what to look for

Research Method

- ▶ Data:
 - ▶ NYSE, AMEX, NASDAQ stocks with CRSP returns and COMPUSTAT accounting data
 - ▶ Sample period: 1963-1990 (as available)
 - ▶ Non-financial firms only (leverage has different meaning for financials)
- ▶ Key Variables:
 - ▶ Market beta (β) – to be estimated
 - ▶ Size (ME) – market equity
 - ▶ Book-to-market equity (BE/ME)
 - ▶ Leverage (A/ME, A/BE)
 - ▶ Earnings-price ratio (E/P)

Research Method - The Critical Innovation

The Double-Sort Methodology

- ▶ "To allow for variation in β that is unrelated to size, we will:
 1. First, sort stocks into 10 portfolios based on **size (ME)**
 2. Then, within each size decile, sort into 10 portfolios based on **pre-ranking β**
- ▶ This creates 100 portfolios with independent variation in size and β , allowing us to separate their effects."

Key to Their Contribution!

Research Method – Double Sort

June of year t :

Step 1: Sort ALL stocks into 10 SIZE portfolios (using NYSE breakpoints)

[Smallest] [2] [3] [4] [5] [6] [7] [8] [9] [Largest]

Step 2: Within EACH size group, sort into 10 beta portfolios

Size Group 1 (Smallest): [Low beta] [2] [3] [4] [5] [6] [7] [8] [9] [High beta]

Size Group 2:

...

Size Group 10 (Largest): [Low beta] [2] [3] [4] [5] [6] [7] [8] [9] [High beta]

Result: 100 portfolios (10×10)

July t to June $t+1$:

Calculate equal-weighted monthly returns for each portfolio

Research Method – Double Sort

	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	β -9	High- β
Panel A: Average Monthly Returns (in Percent)											
All	1.25	1.34	1.29	1.36	1.31	1.33	1.28	1.24	1.21	1.25	1.14
Small-ME	1.52	1.71	1.57	1.79	1.61	1.50	1.50	1.37	1.63	1.50	1.42
ME-2	1.29	1.25	1.42	1.36	1.39	1.65	1.61	1.37	1.31	1.34	1.11
ME-3	1.24	1.12	1.31	1.17	1.70	1.29	1.10	1.31	1.36	1.26	0.76
ME-4	1.25	1.27	1.13	1.54	1.06	1.34	1.06	1.41	1.17	1.35	0.98
ME-5	1.29	1.34	1.42	1.39	1.48	1.42	1.18	1.13	1.27	1.18	1.08
ME-6	1.17	1.08	1.53	1.27	1.15	1.20	1.21	1.18	1.04	1.07	1.02
ME-7	1.07	0.95	1.21	1.26	1.09	1.18	1.11	1.24	0.62	1.32	0.76
ME-8	1.10	1.09	1.05	1.37	1.20	1.27	0.98	1.18	1.02	1.01	0.94
ME-9	0.95	0.98	0.88	1.02	1.14	1.07	1.23	0.94	0.82	0.88	0.59
Large-ME	0.89	1.01	0.93	1.10	0.94	0.93	0.89	1.03	0.71	0.74	0.56

Table I in Fama & French (1992)

Research Method – Double Sort

	Book-to-Market Portfolios										
	All	Low	2	3	4	5	6	7	8	9	High
All	1.23	0.64	0.98	1.06	1.17	1.24	1.26	1.39	1.40	1.50	1.63
Small-ME	1.47	0.70	1.14	1.20	1.43	1.56	1.51	1.70	1.71	1.82	1.92
ME-2	1.22	0.43	1.05	0.96	1.19	1.33	1.19	1.58	1.28	1.43	1.79
ME-3	1.22	0.56	0.88	1.23	0.95	1.36	1.30	1.30	1.40	1.54	1.60
ME-4	1.19	0.39	0.72	1.06	1.36	1.13	1.21	1.34	1.59	1.51	1.47
ME-5	1.24	0.88	0.65	1.08	1.47	1.13	1.43	1.44	1.26	1.52	1.49
ME-6	1.15	0.70	0.98	1.14	1.23	0.94	1.27	1.19	1.19	1.24	1.50
ME-7	1.07	0.95	1.00	0.99	0.83	0.99	1.13	0.99	1.16	1.10	1.47
ME-8	1.08	0.66	1.13	0.91	0.95	0.99	1.01	1.15	1.05	1.29	1.55
ME-9	0.95	0.44	0.89	0.92	1.00	1.05	0.93	0.82	1.11	1.04	1.22
Large-ME	0.89	0.93	0.88	0.84	0.71	0.79	0.83	0.81	0.96	0.97	1.18

Table V in Fama & French (1992)

Research Method - The Critical Innovation

Fama-MacBeth Regressions

- ▶ "We will use the Fama-MacBeth (1973) cross-sectional regression approach:
 1. Each month, run cross-sectional regressions of returns on the variables of interest
 2. Compute time-series averages of the monthly slopes
 3. Use standard errors from the time series to test significance
- ▶ This allows us to test which variables have **non-zero expected premiums**."

Research Method – Key Design Choices

Choice	Their Decision	Why
Sample period	1963-1990	COMPUSTAT data reliable after 1962
Firm exclusion	No financials	Leverage has different meaning
Beta estimation	Portfolio betas assigned to stocks	Individual stock betas too noisy
Timing	Accounting data from t-1 matched with returns from July t to June t+1	Ensures accounting data known before returns

Key References

► Hypothetical Bibliography

- Banz, R.W. (1981). "The relationship between return and market value of common stocks." *Journal of Financial Economics*.
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- Bhandari, L.C. (1988). "Debt/Equity ratio and expected common stock returns." *Journal of Finance*.
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- Fama, E.F. and MacBeth, J. (1973). "Risk, return and equilibrium: Empirical tests." *Journal of Political Economy*.
- Rosenberg, B., Reid, K., and Lanstein, R. (1985). "Persuasive evidence of market inefficiency." *Journal of Portfolio Management*.
- Sharpe, W.F. (1964). "Capital asset prices." *Journal of Finance*.
- Stattman, D. (1980). "Book values and stock returns." *The Chicago MBA*.

One Page Summary

Section	Summary
Problem	CAPM says beta explains returns, but anomalies literature suggests size, value, leverage, and E/P also matter. Which are real?
Expected Results	Beta will have little power; key factors will absorb the others.
Method	Double-sort portfolios to separate key factor and beta; Fama-MacBeth regressions on individual stocks with portfolio betas. Data: CRSP + COMPUSTAT, 1963-1990, non-financials.
Key References	Sharpe (1964), Banz (1981), Basu (1983), Bhandari (1988), Rosenberg et al. (1985), Fama-MacBeth (1973).

What Made Their Proposal Compelling?

- ▶ **Important question:** Challenges the foundation of asset pricing
- ▶ **Clear hypothesis:** Many variables are redundant
- ▶ **Innovative method:** Double sorting to separate correlated effects
- ▶ **Feasible:** Data exists, methods are established
- ▶ **High impact potential:** If correct, would change how everyone thinks about risk and return

Where to Get Ideas?

- ▶ Not always from Journals!
- ▶ Varian's Advice: Look for ideas in the **world**, not in journals
- ▶ "In my experience journals really aren't a very good source of original ideas. You can get lots of things from journal articles—technique, insight, even truth. But most of the time you will only get someone else's ideas."
— Hal Varian

Sources of Ideas

Source	Examples
Newspapers	Wall Street Journal, Financial Times, NY Times, Caixin
Magazines	The Economist, Bloomberg Businessweek, Caixin
Conversations	Talk to people in business, practitioners
Podcast	Economic news, policy debates
Your own life	Varian's "A Model of Sales" came from buying a TV

- Things that seem puzzling
- Things that don't match theory
- "What's wrong with this picture?"

Varian's TV Story

- ▶ A Classic Example: "A Model of Sales"

"I had decided to get a new TV so I followed the ads in the newspaper to get an idea of how much it would cost. I noticed that the prices fluctuated quite a bit from week to week. It occurred to me that the challenge to economics was not why the prices were sometimes low (i.e., during sales) but why they were ever high."

- ▶ **The insight:** Two types of consumers—informed (read ads) and uninformed (don't read ads). Stores use sales to price discriminate.
- ▶ **The result:** A published paper + a good deal on the TV!

Is Your Idea Worth Pursuing?- The "Ah! So that explains it!" Test

- ▶ Can I explain this idea to a **non-economist**?
- ▶ Is it **interesting**? (If not, no one cares if it's correct)
- ▶ What would follow if this idea is correct?
- ▶ What's the **opportunity cost** of working on this idea?

When to Look at the Literature?

- ▶ Varian's Advice: Don't Look Too Soon!
- ▶ Wait a few weeks before doing a systematic literature review
 - ▶ Practice: You need the experience of developing a model
 - ▶ Originality: If you read others first, your thinking will be shaped by their views
 - ▶ Incubation: Ideas need time to develop—start modeling early

"If you're very lucky and your model is not in the literature, maybe you're wrong. Here is where the advisor is really important." — Varian

When to Look at the Literature?

- ▶ But if you work on empirics
- ▶ Check early!
- ▶ You need to know what's already been tested!
- ▶ But still, **Don't Let Existing Literature Constrain Your Thinking!**
- ▶ Spend **a few days** thinking about your idea and sketching a research design before diving deep into the literature.

MUST DOs for an Empirical Idea

- ▶ Check **data availability** early.
- ▶ **Scan on key papers** to ensure your question hasn't been answered?
 - ▶ But sometime, not the end of the story
- ▶ **Think independently first**—spend a few days developing your own framing before diving deep
- ▶ **Talk to advisors early**—they can save you from dead ends
- ▶ **Don't do a full literature review immediately**—that can wait until you have a promising research design

Where to Find Idea - Practical Suggestions for Beginners

- ▶ Skim conference websites (AFA, WFA, NBER, EFA, CICF) from recent 2-3 years
- ▶ Check journal homepages (JF, RFS, JFE, JFQA)
- ▶ Follow researchers you love (personal website)
- ▶ Then ask:
 - ▶ Can I extend this to new settings? New markets, new products, natural experiments
 - ▶ Can I supply a new explanation? New variable, privileged data access
 - ▶ Can I improve the method? Technical background advantage

Idea Checklist

- ▶ Is the question **important**?
- ▶ Is the setting **distinctive**?
- ▶ Is the data **hard to compile**?
- ▶ Are there **persuasive ex-ante** reasons for the new ingredients to work?

Proposal Template – Adapted from Fama-French(1992)

Section	What to Write
1. Title	Descriptive, specific
2. Research Question	One sentence: "This paper asks..."
3. Motivation	Why should anyone care? Gap between theory and reality?
4. Expected Contribution	What will we learn that's new?
5. Literature Review	Key papers (5-10), what's missing
6. Methodology	Data, sample, empirical strategy/model
7. Expected Results	What do you hypothesize?
8. Timeline	Milestones and deadlines
9. References	Complete citations

Proposal Checklist

Before you show your advisor

- ▶ Research question is clear and specific
- ▶ Motivation explains why anyone should care
- ▶ Contribution is clearly stated (what's new?)
- ▶ Literature review covers key papers (5-10)
- ▶ Gap in literature is identified
- ▶ Methodology is feasible (data exists!)
- ▶ Expected results are outlined
- ▶ Timeline is realistic
- ▶ References are complete and properly formatted

Common Proposal Mistakes to Avoid

Mistake	Fama-French Counterexample
Question too vague	“Which variables explain stock returns?” → Specific: “Do some key factors capture all anomalies?”
No clear contribution	They explicitly state: "Our main result is that size and BE/ME capture the roles of leverage and E/P."
Data not checked	They knew CRSP and COMPUSTAT existed and were mergeable
Method beyond skills	Fama-MacBeth was established; double sorting was their innovation
Literature as annotated bibliography	They synthesized prior work into a coherent puzzle

Getting Feedback

Person	What They Can Tell You
Advisor	Is this worth doing? Any gaps?
Other faculty	Different perspective, literature you missed
Fellow students	Is it clear to a non-expert?
Practitioners	Does it match real-world experience?

"Talk to your advisor, talk to your fellow students, talk to your wife, husband, girlfriend, boyfriend, neighbor, or pet... whoever you can get to listen." — Varian

The Golden Rules

1. **Start with a puzzle** – Something that doesn't fit existing theory
2. **Ask a clear question** – Can you state it in one sentence?
3. **Design a test that can reject your hypothesis** – Don't just confirm what you believe
4. **Write a proposal first** – Clarify your thinking, get feedback early
5. **Check data availability** – Before you fall in love with an idea
6. **Be parsimonious** – The simplest model that captures the essence